

Ananmay Som Singh

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EDUCATION

University of Maryland, College Park | College Park, MD

Expected May 2028

Bachelor of Science in Computer Engineering

Relevant Coursework: Algorithms, Data Structures, Computer Systems, Compilers, Probability & Statistics

EXPERIENCE

GSAlpha Labs | Real Estate Transaction Automation

May 2026 – Present

Software Engineering Intern · Part-time during academic year

San Francisco, CA

- Built HomeFlow AI, a Next.js/TypeScript platform in production that files closing packets, drafts offers, and tracks deadlines for California agents.
- Cut purchase-agreement data extraction from 35 minutes of manual entry to under 30 seconds with an LLM pipeline that reached 96% field accuracy on 15 real transactions; a human reviews before filing.
- Shipped an offer workflow that drafts a purchase agreement from a 175-page disclosure packet in under 5 minutes, flagging only conflicting fields.
- Improved document classification from 14/27 to 27/27 correct on a real closing packet by matching each form's unique footer text.
- Automated contingency deadlines across California business days and holidays, showing the rule behind each date so agents trust it.

theconviction.ai | Financial Research Automation

May 2025 – Aug 2025

Software Engineering Intern

Remote

- Replaced 12 hours a week of manual data collection with a FastAPI/PostgreSQL pipeline that pulls filings, transcripts, and news for 50+ companies.
- Added timeouts, retries, and deduplication to handle unreliable sources and overlapping feeds, keeping one clean record per company.
- Shipped a Next.js research tool used daily by 4 analysts that links every finding on a company timeline to its source filing.

SEDS at UMD | SatFab CubeSat GPS Team

Sep 2024 – Feb 2026

Software Engineer

College Park, MD

- Built the Python test framework for 26 Verilog modules in a CubeSat GPS receiver; adopted as the team standard after training 6 engineers.
- Cut hardware regression time from 4 hours to 95 minutes by running testbenches in parallel, so it runs before every merge instead of overnight.
- Automated waveform checking and pass/fail reporting so integration bugs were caught in simulation instead of on the hardware.

PROJECTS

Tape: Market Data Capture & Replay | Go, AWS S3, ECS, CloudWatch, Terraform, Docker

AnanmayS/tape

- Built a Go service that records live exchange feeds to S3 and replays them byte-for-byte, sustaining 49.8k messages/sec in load tests.
- Detected all 3 feed gaps (3,240 lost messages) in fault-injection tests and flagged the affected windows before any backtest runs on them.
- Reduced feed-storage size by a factor of 5.2 using delta-encoded columnar storage.

Wildebeest: Distributed Wildlife Photo Pipeline | TypeScript, Node.js, Python, PostgreSQL, Redis, Docker

AnanmayS/wildebeest

- Built a distributed pipeline that sorts wildlife camera-trap photos with 2 AI models across many machines; 1,000 real photos ran with 0 failures.
- Built the scheduler on Postgres and Redis that lost or double-counted 0 results across 30 injected crashes by reassigning dead workers' tasks.
- Cut worker crash recovery from 5.6 s to 0.16 s by detecting dead workers instantly through Docker events instead of missed check-ins.

ShowdownRL: Reinforcement Learning Agent | Python, PyTorch, PPO, Optuna, Playwright, Gymnasium

AnanmayS/ShowdownRL

- Trained a PPO agent for Pokémon Showdown that won 79% of 1,000 simulated battles against a type-aware scripted bot, masking illegal moves.
- Built a tuning pipeline with Optuna that promotes a new model only if it beats the baseline on every metric across multiple test seeds.
- Drove real matches through the browser with Playwright and saved every battle log, so any reported win rate traces back to the games behind it.

TECHNICAL SKILLS

Languages: Go, Python, Java, C++, C, TypeScript, JavaScript, SQL, Verilog

Backend & Data: FastAPI, Node.js, PostgreSQL, Redis, Drizzle, SQLAlchemy, Supabase, Zod, WebSockets, Concurrency

Infrastructure & Tooling: AWS, S3, ECS, CloudWatch, Terraform, Docker, Linux, GitHub Actions (CI/CD), Vercel, pytest, Vitest

Frameworks & ML: React, Next.js, PyTorch, LLM extraction and evaluation